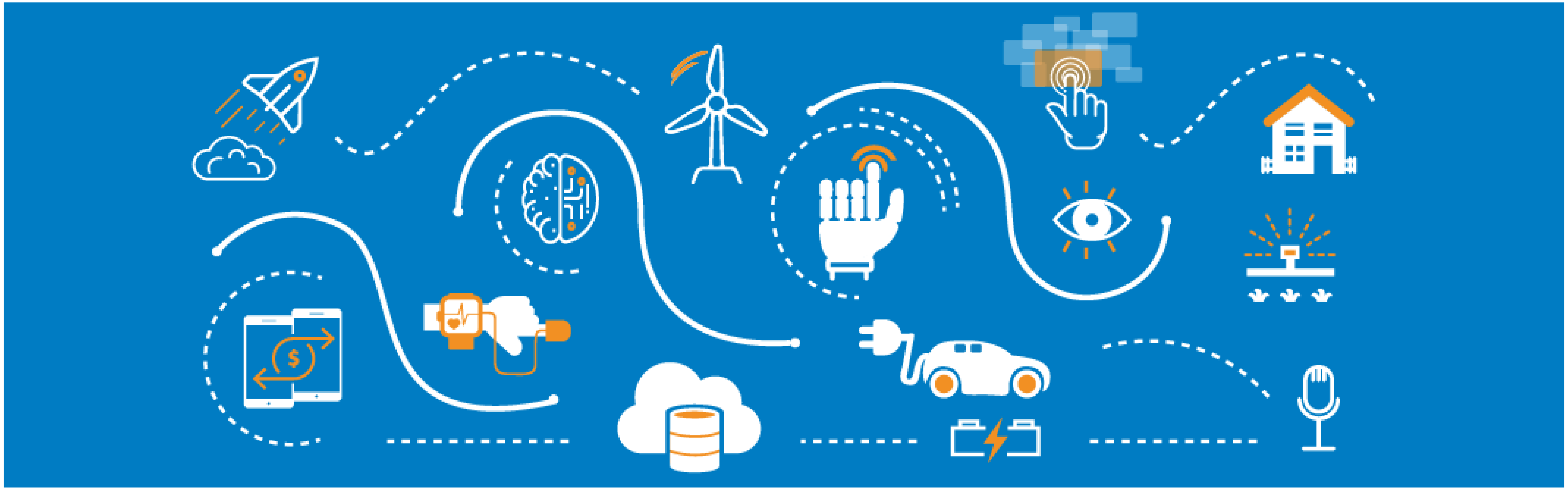


Model Evaluation and Interpretability

Kien Tran

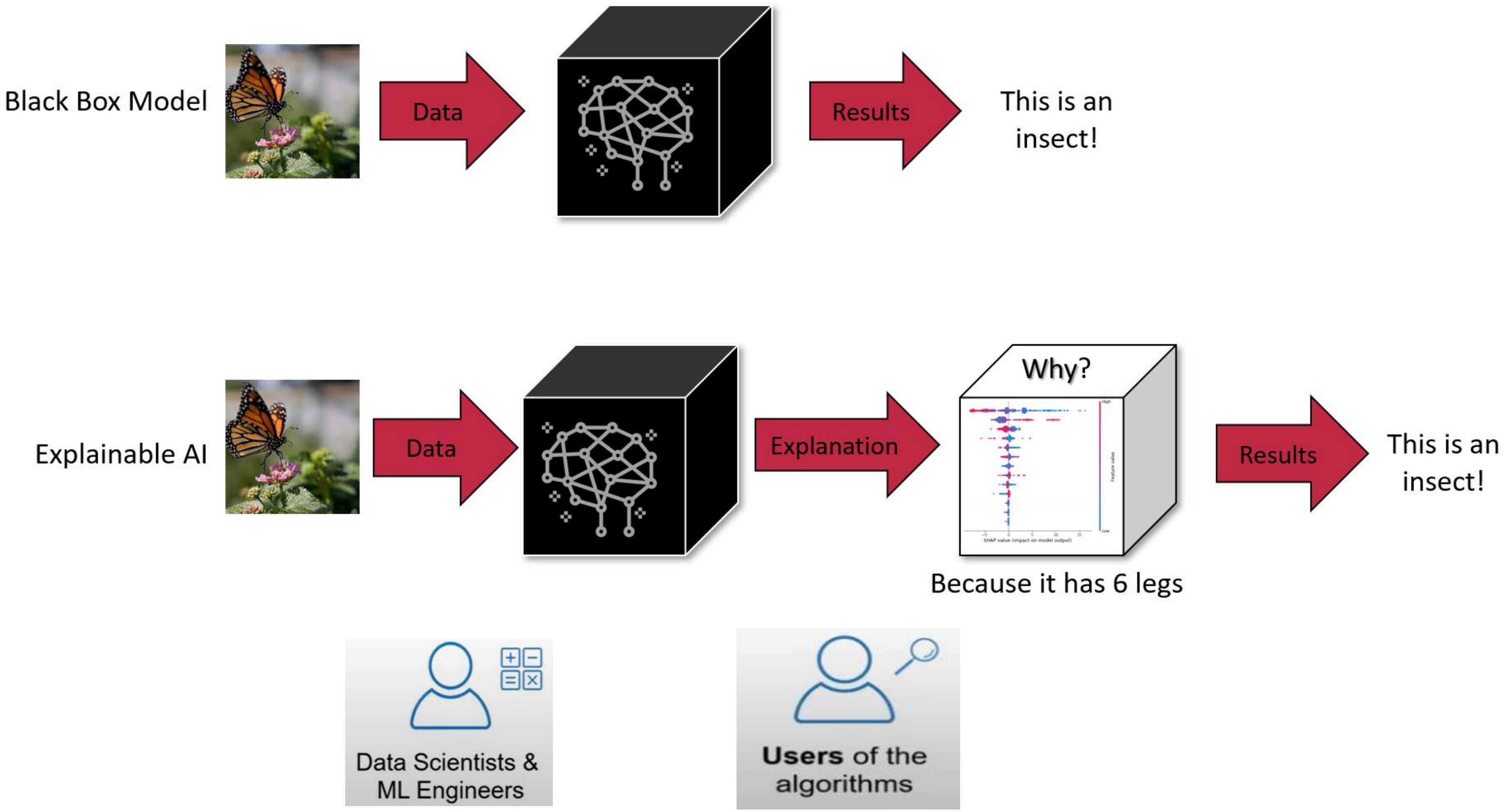
Table of Contents

01 Metrics for evaluating explainability

- 1.1 Introduction
- 1.2 Accuracy, Precision
- 1.3 Recall, F1-score
- 1.4 ROC, AUC

1. What is explainable AI?

1. What is Explainable AI?



2. Why Explainable AI?

ABILITY

- **AI < Human (can not deploy)**

⇒ Identify root cause of failure AI

⇒ Clear improvement model

- **AI = Human**

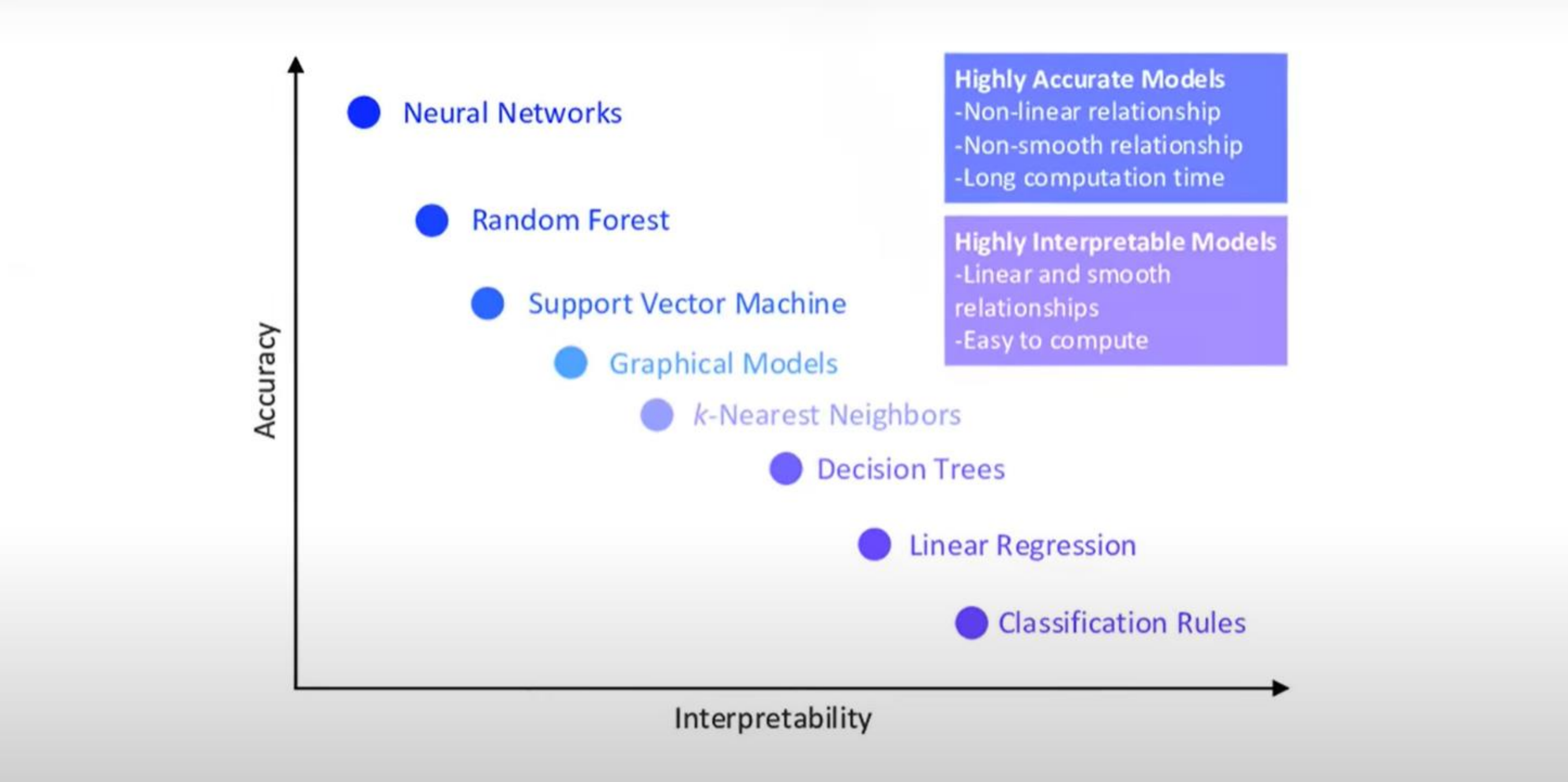
⇒ Can build trust for model with users

- **AI > Human**

⇒ Can use model to guide human that task

1.1. Introduction

3. Overview Explainable AI

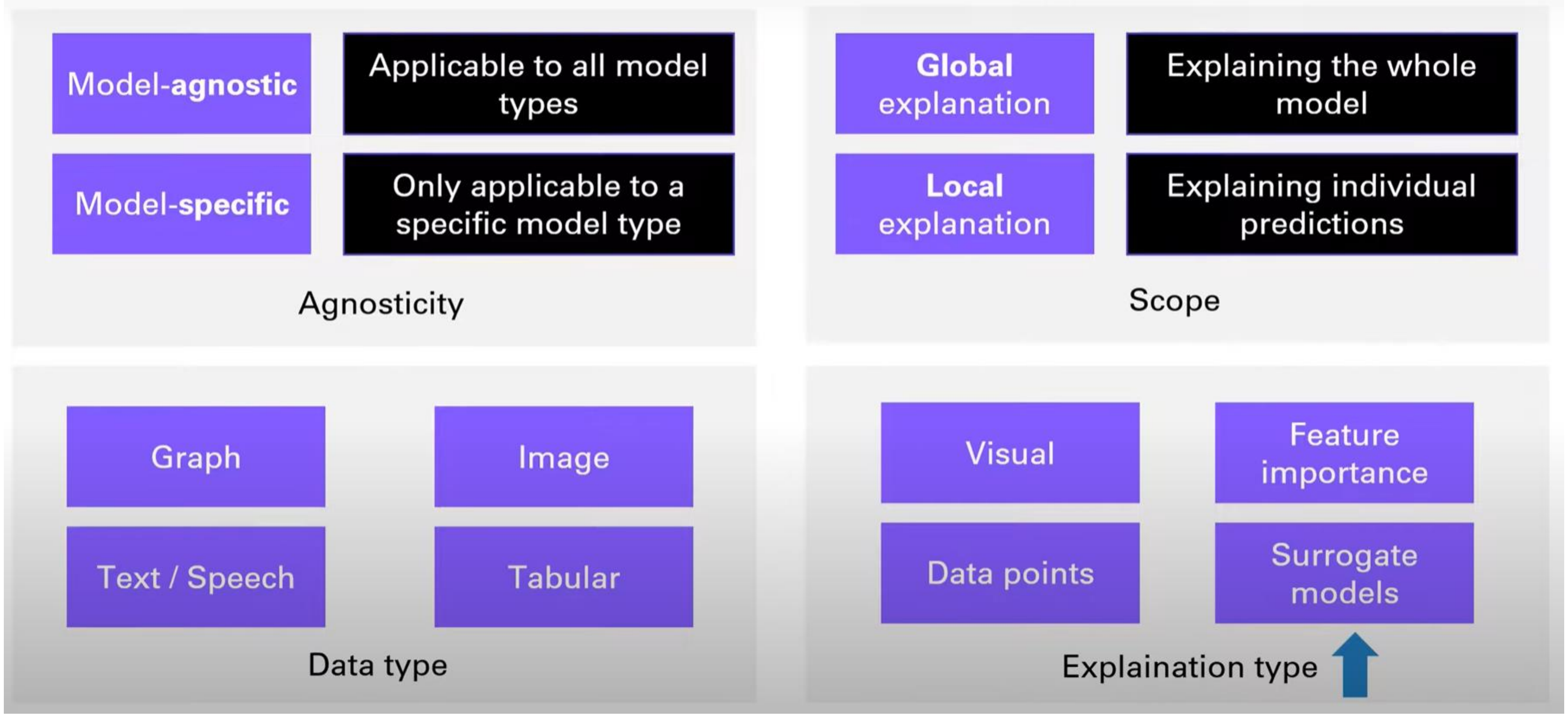


Source: Machine Learning for 5G/B5G Mobile and Wireless Communications: Potential, Limitations, and Future Directions

3. Overview Explainable AI

2 options to understand models

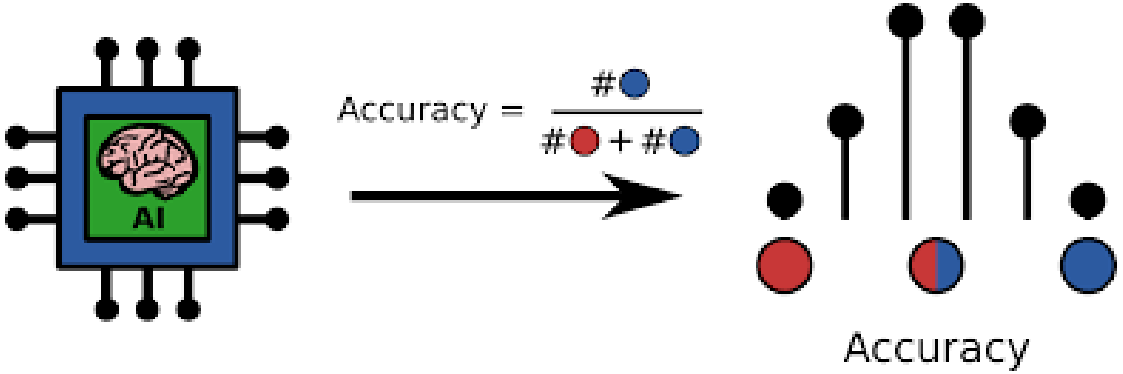
4. Categorization in XAI



1.2. Classification Metrics

1. Accuracy

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$




```python
from __future__ import print_function
import numpy as np

def acc(y_true, y_pred):
    correct = np.sum(y_true == y_pred)
    return float(correct)/y_true.shape[0]

y_true = np.array([0, 0, 0, 0, 1, 1, 1, 2, 2, 2])
y_pred = np.array([0, 1, 0, 2, 1, 1, 0, 2, 1, 2])
print('accuracy = ', acc(y_true, y_pred))
```

```
accuracy =  0.6
```

Scratch compute

```python
from sklearn.metrics import accuracy_score
print('accuracy = ',accuracy_score(y_true, y_pred))
```

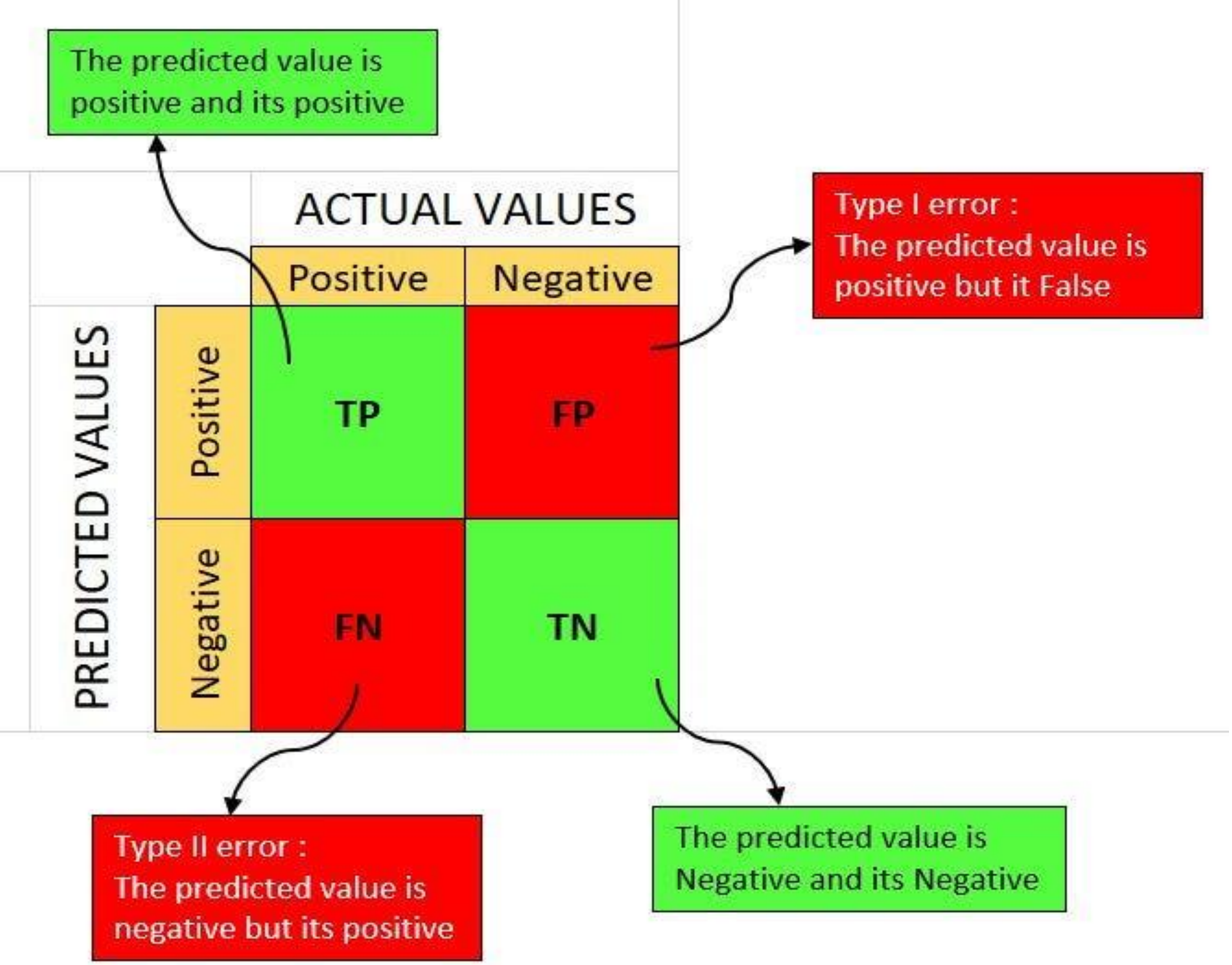
```
accuracy =  0.6
```

Library compute

1.2. Classification Metrics

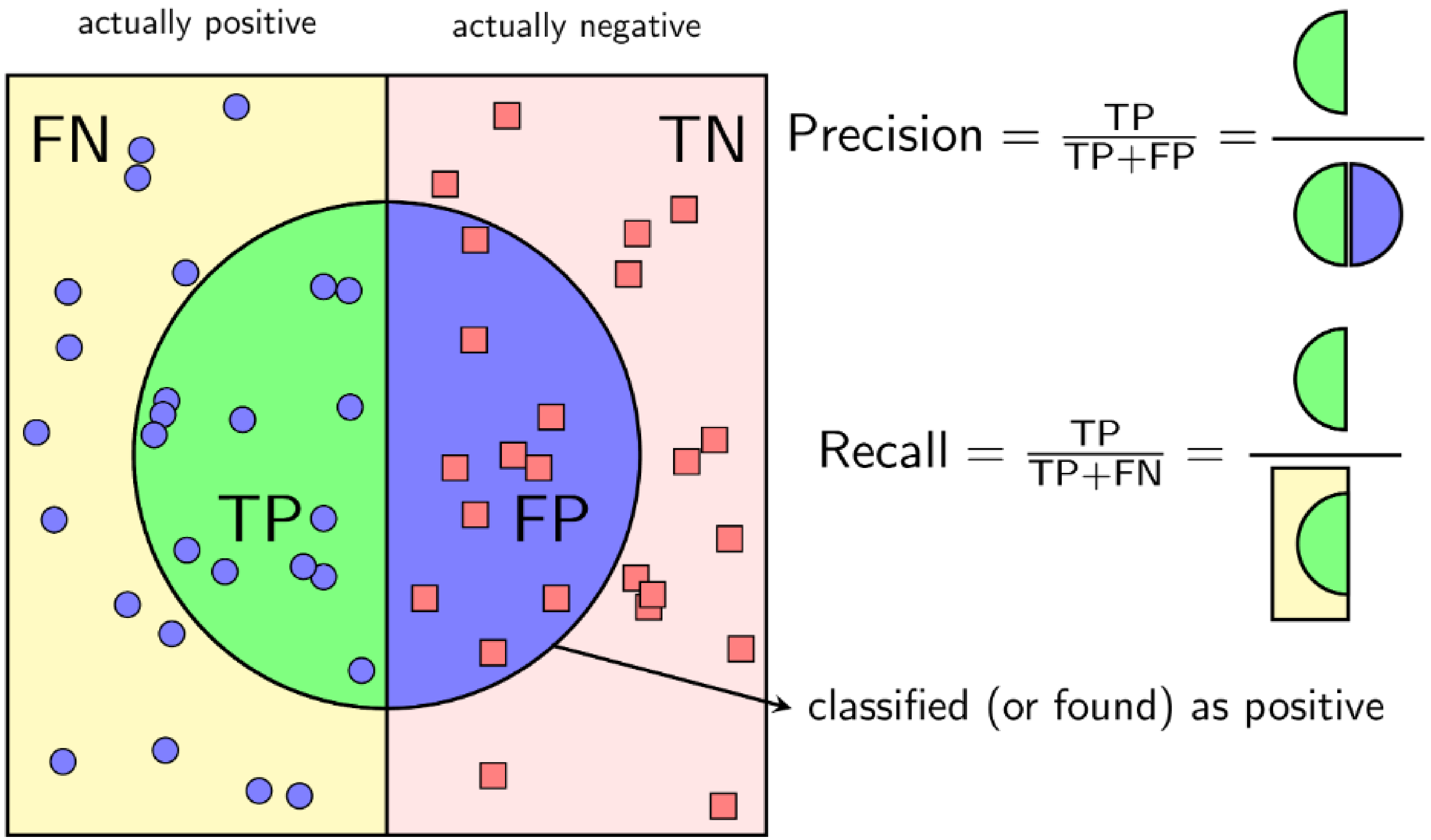
2. Confusion Matrix

- **True Positives (TP):** When you correctly guess something belongs to a group, and it actually does.
- **True Negatives (TN):** When you correctly guess something doesn't belong to a group, and it actually doesn't.
- **False Positives (FP):** When you mistakenly guess something belongs to a group, but it actually doesn't.
- **False Negatives (FN):** When you mistakenly guess something doesn't belong to a group, but it actually does.



1.2. Classification Metrics

3. Precision & Recall



1.2. Classification Metrics

3. Precision & Recall

Giả sử chúng ta có một mô hình học máy để phân loại email thành spam hoặc không spam. Dữ liệu tập huấn gồm 100 email, trong đó:

•60 email là spam (lớp dương)

•40 email không phải spam (lớp âm)

Mô hình dự đoán 70 email là spam và 30 email không phải spam.

Ma trận nhầm lẫn (confusion matrix) cho mô hình này như sau:

Dự đoán	Spam	Không spam
Spam	**TP = 42**	FP = 28
Không spam	FN = 18	**TN = 12**

Precision (Độ chính xác) là tỷ lệ các email được dự đoán là spam thực sự là spam:

Precision = TP / (TP + FP) = 42 / (42 + 28) = 0.6

Recall (Độ thu hồi) là tỷ lệ các email thực sự là spam được dự đoán là spam:

Recall = TP / (TP + FN) = 42 / (42 + 18) = 0.7

•**Precision cao** cho biết mô hình ít dự đoán sai email không phải spam là spam.

•**Recall cao** cho biết mô hình ít bỏ sót các email spam.

Lựa chọn giữa precision và recall phụ thuộc vào mục đích của mô hình.

•**Nếu mục tiêu** là giảm thiểu số lượng email không phải spam được dự đoán là spam, ta nên ưu tiên precision.

•**Nếu mục tiêu** là giảm thiểu số lượng email spam bị bỏ sót, ta nên ưu tiên recall.

1.2. Classification Metrics

4. F1 Score

A high F1-score indicates that the model has a good balance between precision and recall.

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Example: Mail Spam

•**Mô hình A** có precision = 0.8 và recall = 0.7, F1-score = 0.75.

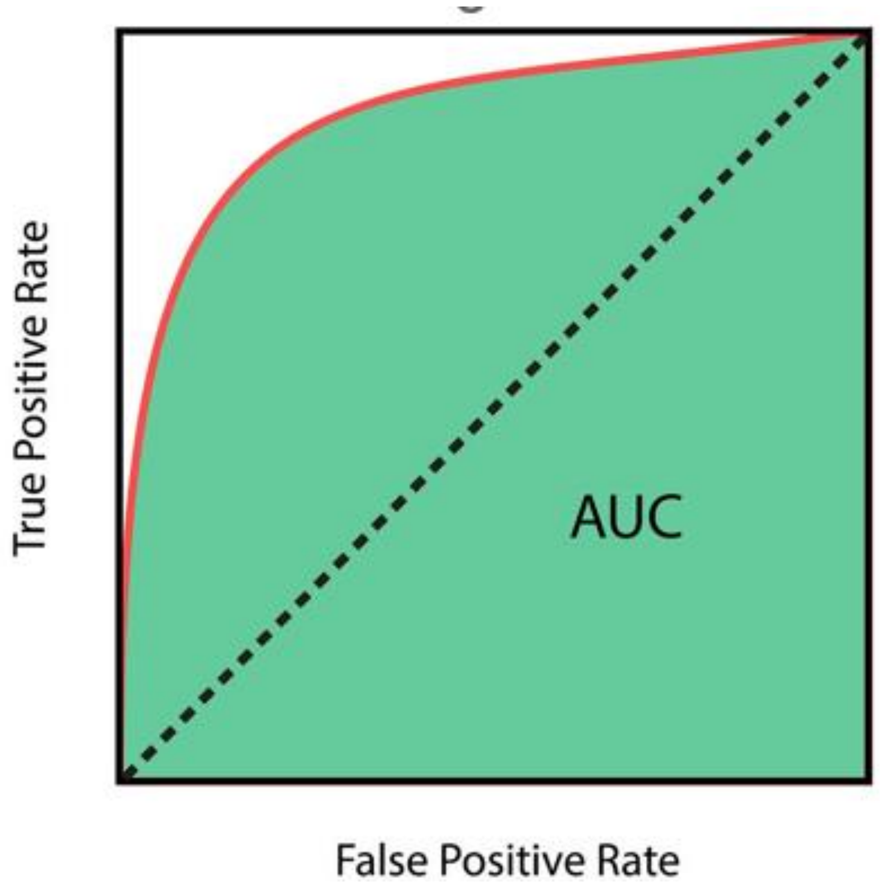
•**Mô hình B** có precision = 0.7 và recall = 0.8, F1-score = 0.75.

Cả hai mô hình A và B đều có F1-score là 0.75, cho thấy hiệu suất tương đương. Tuy nhiên, mô hình A có precision cao hơn, nghĩa là nó ít dự đoán sai email không phải spam là spam hơn.

1.2. Classification Metrics

5. ROC & AUC

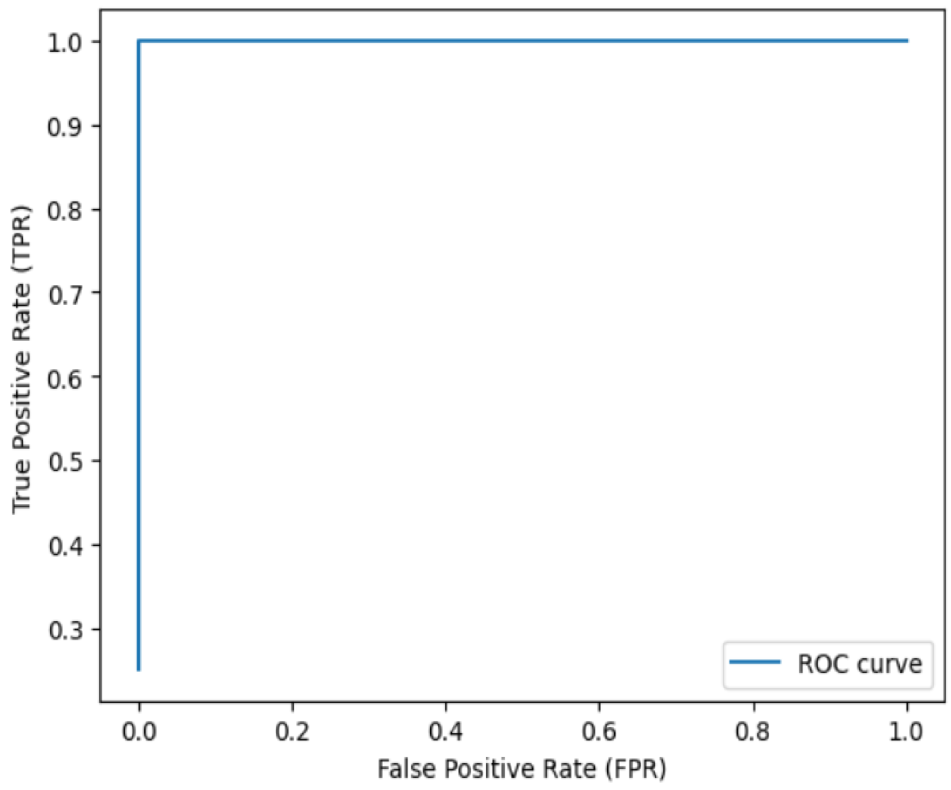
- ROC is a curve that represents the ability of a machine learning model to distinguish between classes.
- AUC is the area under the ROC curve. A high AUC indicates that the model has good ability to discriminate between classes.



1.2. Classification Metrics

5. ROC & AUC

```python
import numpy as np
import matplotlib.pyplot as plt

# Điểm xác suất
scores = np.array([0.9, 0.8, 0.7, 0.6, 0.5, 0.4, 0.3, 0.2, 0.1, 0.0])

# Nhãn email
labels = np.array([1, 1, 1, 1, 0, 0, 0, 0, 0, 0])

# Số lượng email
n_spam = np.sum(labels)
n_not_spam = len(labels) - n_spam

# FPR và TPR cho các điểm ngưỡng khác nhau
fpr, tpr = [], []
for threshold in scores:
    tp = np.sum(scores[labels == 1] >= threshold)
    fp = np.sum(scores[labels == 0] >= threshold)
    fpr.append(fp / n_not_spam)
    tpr.append(tp / n_spam)

# Vẽ đường ROC
plt.plot(fpr, tpr, label="ROC curve")
plt.xlabel("False Positive Rate (FPR)")
plt.ylabel("True Positive Rate (TPR)")
plt.legend()
plt.show()

# Tính AUC
auc = np.trapz(tpr, fpr)
print("AUC:", auc)
```

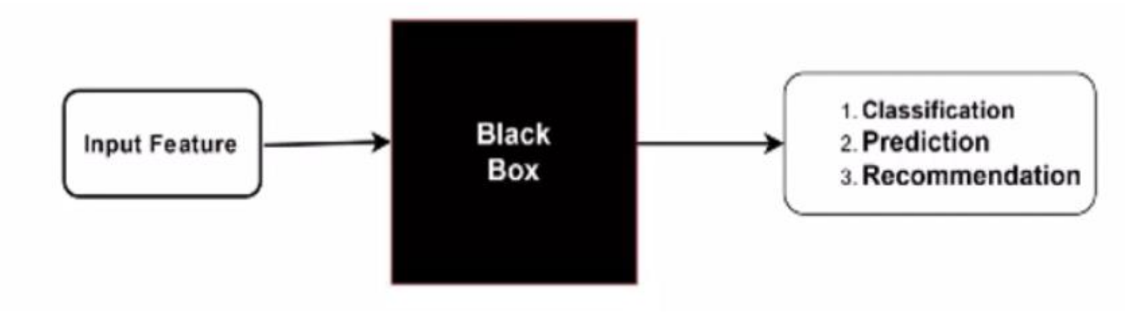



```
AUC: 1.0
```

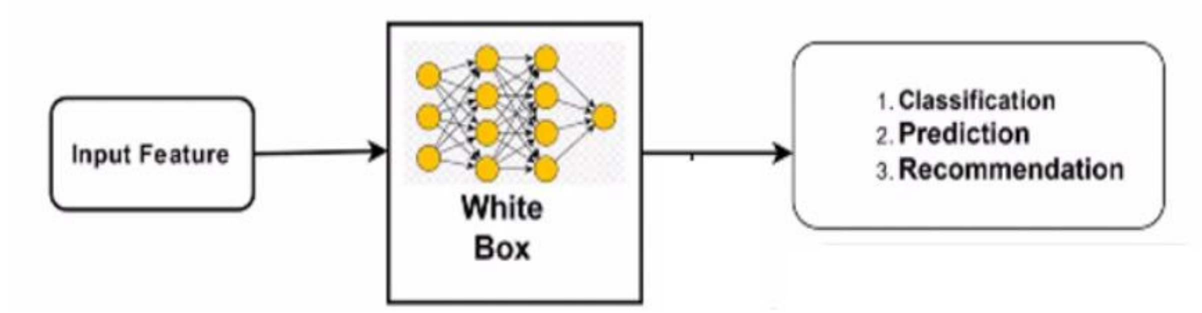
02 Explainability in Machine Learning

- **2.1** Black-box vs White-Box models
- **2.2** Interpretable ML algorithms
- **2.3** Post-hoc explainability techniques

2.1. Black Box vs White Box Models



- Lack of transparency
- Limited Interpretability
- Complex internal representations
- Difficulty in error diagnosis

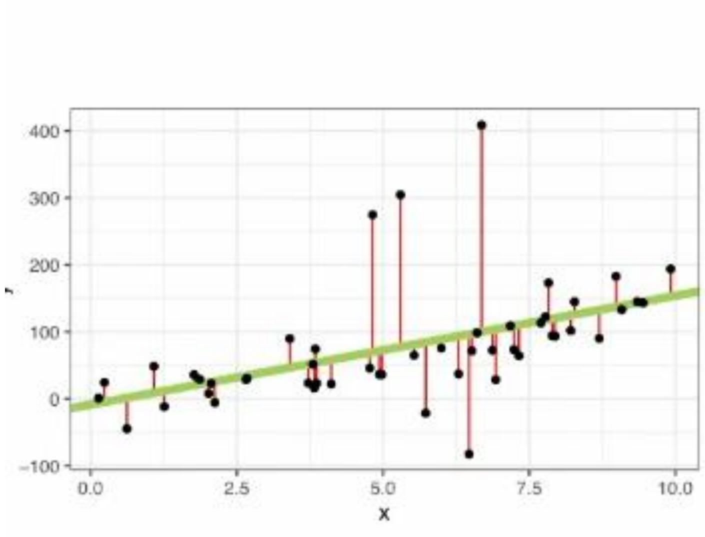


- Transparency
- Interpretable features
- Explainable decisions
- Error Diagnosis and Debugging

2.2. Interpretable ML Algorithms

1. Linear Regression

$$y = w_0 + w_1 x_1 + \ldots w_p x_p + \epsilon,$$



Y: output
W: weight
W0 : intercept
X: input feature
e: error between the predictions and the true outcome

=> methods used to estimate the weights is known as

ordinary least squares (OLS)

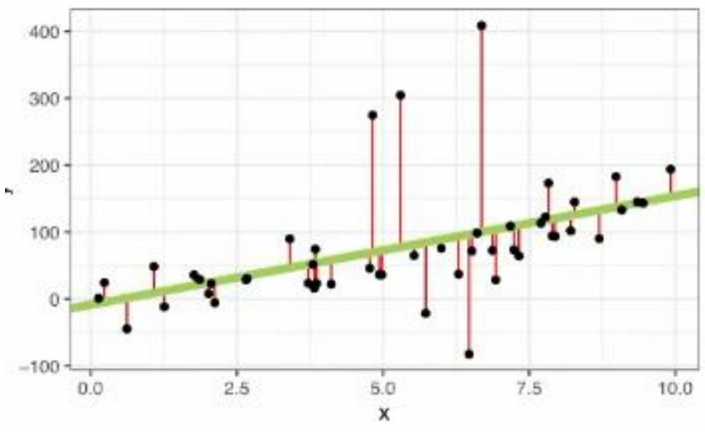
2.2. Interpretable ML Algorithms

1. Linear Regression

Weight: weight indicates the influence an input feature has on the output

R^2: R^2 is an important measure indicating the goodness-of-fit for linear regression models.
It tells us how much of the total variance in the output is explained by the model
0 < R^2 < 1

Feature Importance (t-statistic): *The t-statistic is the estimated weight of an input feature scaled with its standard error (t-statistic = weight/standard error in weight estimate). More important , more weight, mort t-statistic, vice versa*



2.2. Interpretable ML Algorithms

1. Linear Regression

Example: Medical Cost Personal Dataset from Kaggle, to examine the relationship between insurance charges (y) and a number of independent variables such as age, sex, bmi, number of children, smoking and region

Dataset: insurance

Code: linear regression

```
Individual 1:
age          19.0
bmi          27.9
children      0.0
sex           0.0
smoke         1.0
northwest     0.0
southeast     0.0
southwest     1.0
insurance charge:  16884.924

Individual 2:
age          19.00
bmi          32.11
children      0.00
sex           0.00
smoke         0.00
northwest     1.00
southeast     0.00
southwest     0.00
insurance charge: 2130.6759

Individual 3:
age          60.000
bmi          36.005
children      0.000
sex           0.000
smoke         0.000
northwest     0.000
southeast     0.000
southwest     0.000
insurance charges: 13228.84695
```

2.2. Interpretable ML Algorithms

2. Logistic Regression & Decision Trees

Exercise: Diabetes Prediction Dataset from Kaggle, to examine which independent variables (age, gender, bmi, hypertension, heart disease, blood glucose levels, HbA1C levels and smoking history) can help predict diabetes (y)

Dataset: [diabetes]

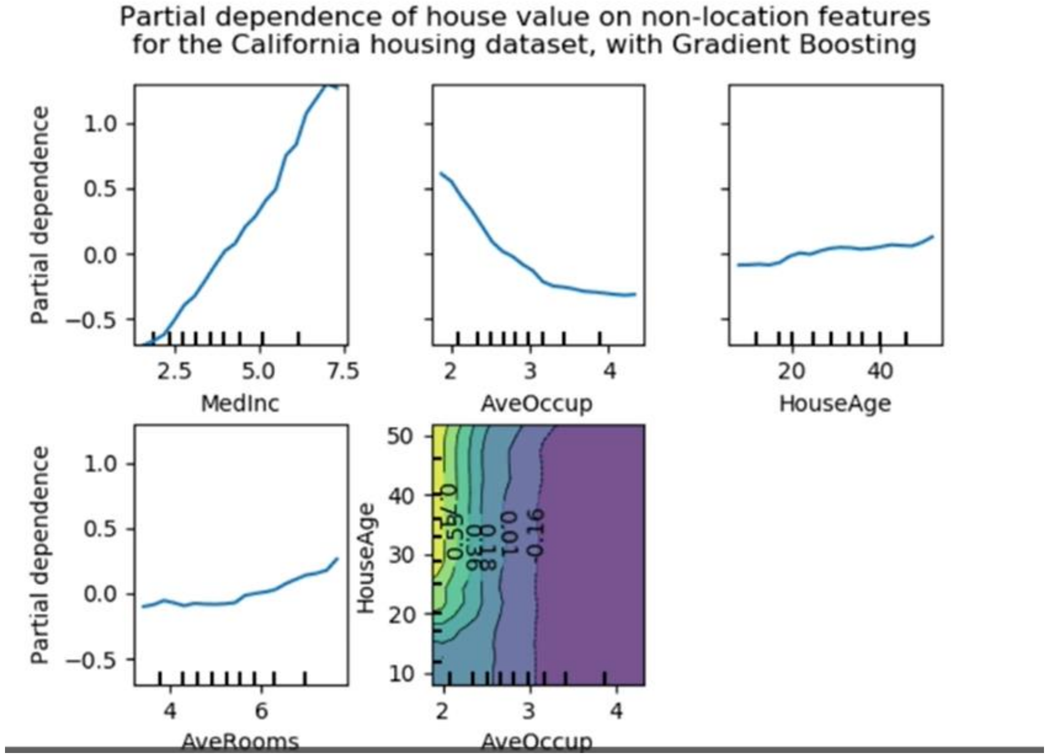
Code: [sample code]

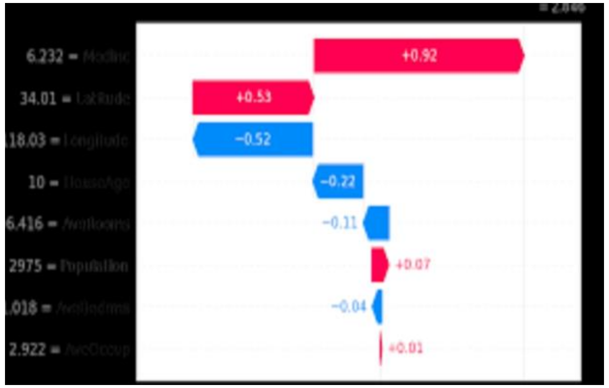
gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
Female	80	0	1	never	25.19	6.6	140	0
Female	54	0	0	No Info	27.32	6.6	80	0
Male	28	0	0	never	27.32	5.7	158	0
Female	36	0	0	current	23.45	5	155	0
Male	76	1	1	current	20.14	4.8	155	0
Female	20	0	0	never	27.32	6.6	85	0
Female	44	0	0	never	19.31	6.5	200	1
Female	79	0	0	No Info	23.86	5.7	85	0
Male	42	0	0	never	33.64	4.8	145	0
Female	32	0	0	never	27.32	5	100	0
Female	53	0	0	never	27.32	6.1	85	0
Female	54	0	0	former	54.7	6	100	0
Female	78	0	0	former	36.05	5	130	0

2.3. Post-hoc expandability techniques

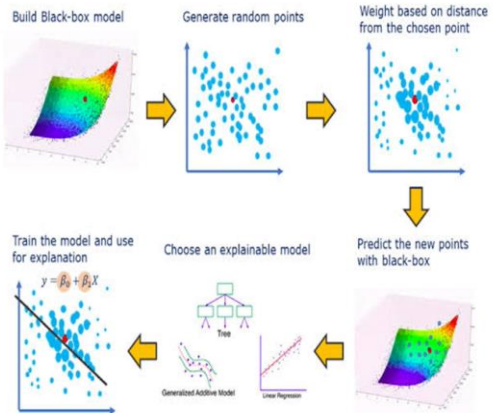
1. Feature importance
2. Partial Dependence Plots (PDP)

3. SHAP Values
4. LIME (Local Interpretable Model-Agnostic Explanation)





SHAP



LIME

03 Model Interpretation Techniques

- 3.1 Microsoft InterpretML
- 3.2 LIME
- 3.3 SHAP

3.1. Microsoft InterpretML

Explain the overall model behavior

Explain an individual prediction

Dataset

[InterpretML]

[Sample Code]

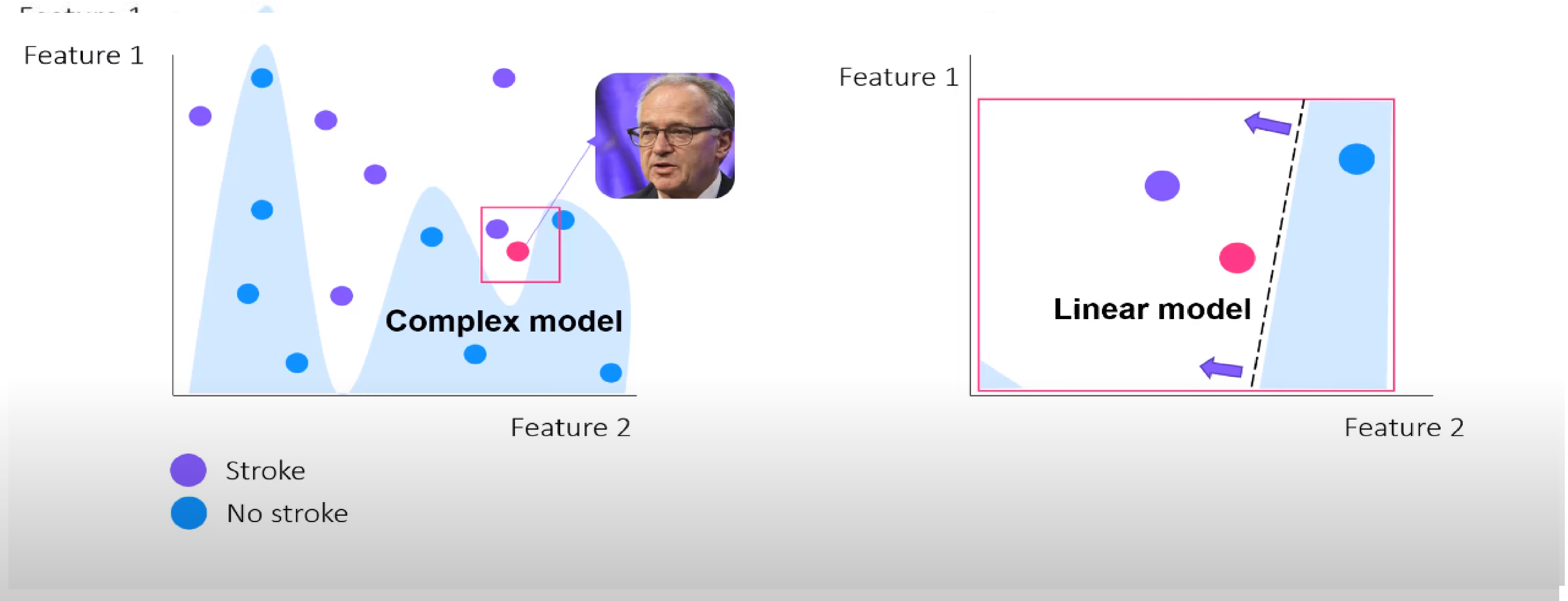
3.1. Microsoft InterpretML

Exercise

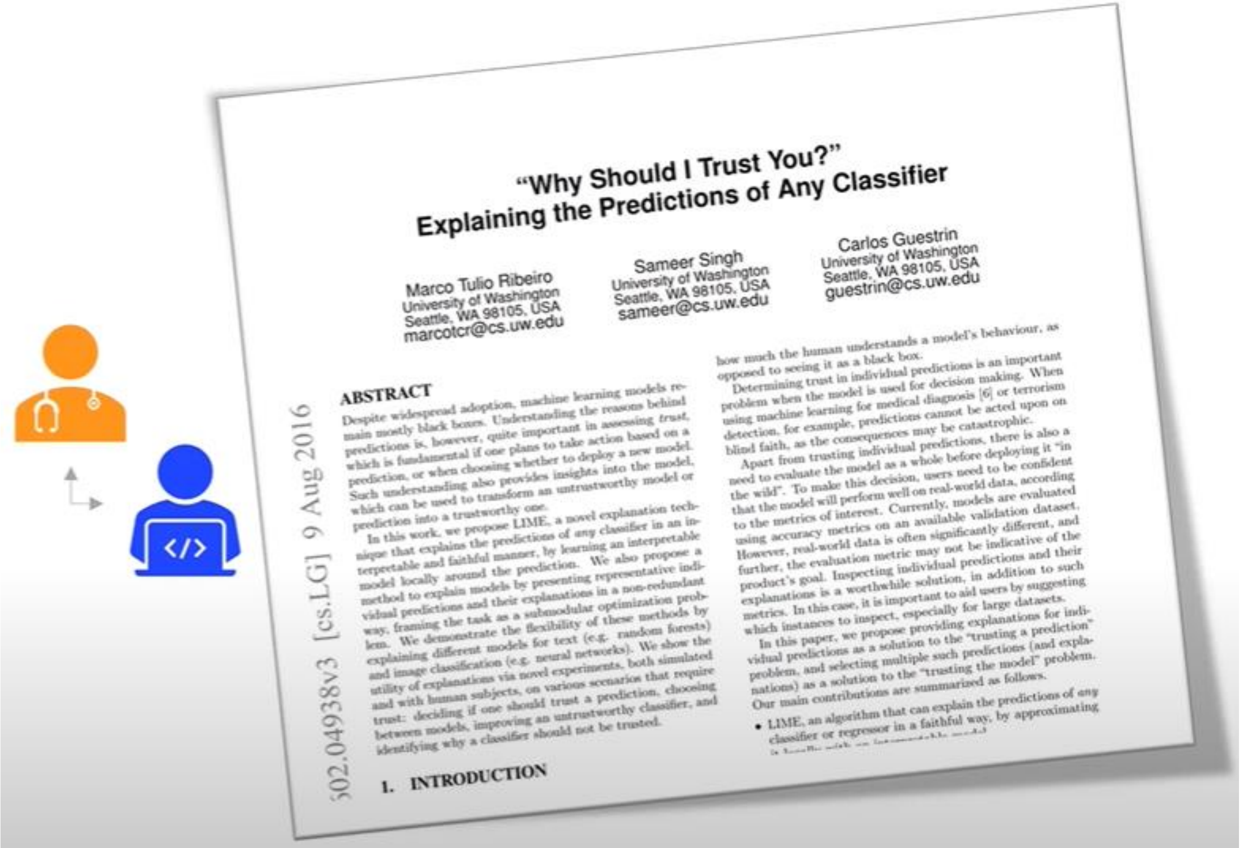
Using InterpretML for Dataset in **Interpretable ML Algorithms Section**

3.2. LIME

LIME (Local Interpretable Model-Agnostic Explanation)



- Work on any black box ML model
- Model internal are hidden
- Works with many data types
- Using prior knowledge we can validate the explanations and create trust
- Explanations are locally faithful, but not necessarily globally



3.2. LIME

The Math in LIME

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \; \overbrace{\mathcal{L}(f, g, \pi_x)}^{\text{Good approximation}} + \overbrace{\Omega(g)}^{\text{Stay simple}}$$

Family of interpretable models

Complex model

Simple interpretable model

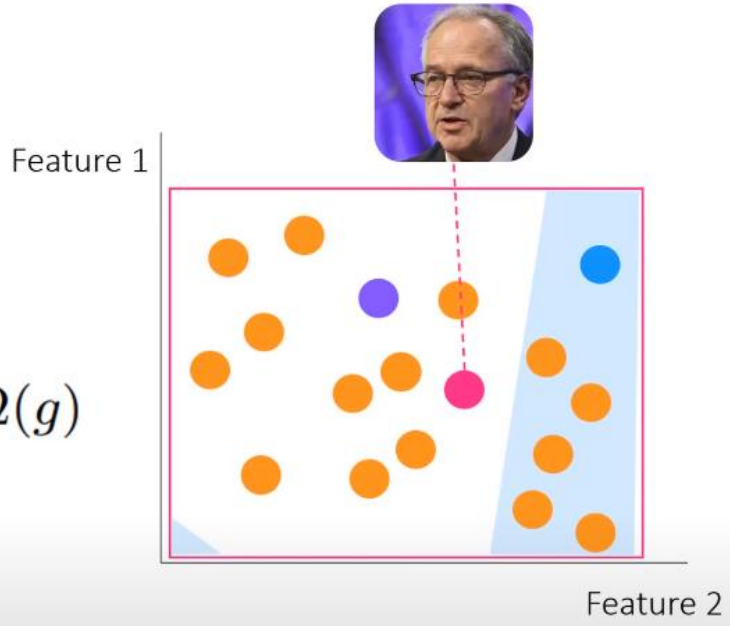
Proximity

Age | Gender | Heart Disease | ...

3.2. LIME

How To Train The Surrogate

$$\xi(x) = \underset{g \in G}{\operatorname{argmin}} \; \boxed{\mathcal{L}(f, g, \pi_x)} + \Omega(g)$$



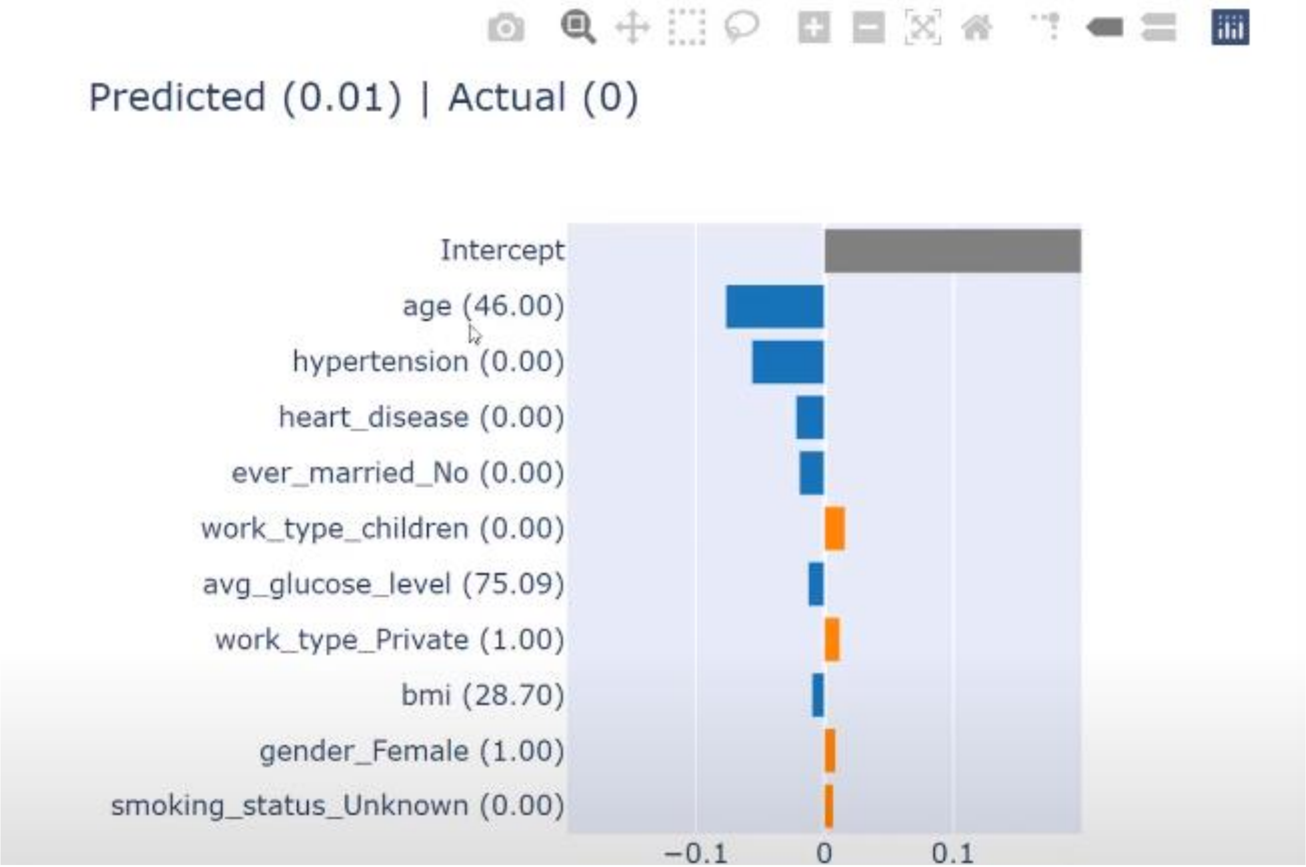
$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in \mathcal{Z}} \pi_x(z) \left(f(z) - g(z')\right)^2$$

New dataset:
- Labels: Prediction of complex model
- Features: Newly generated datapoints

3.2. LIME

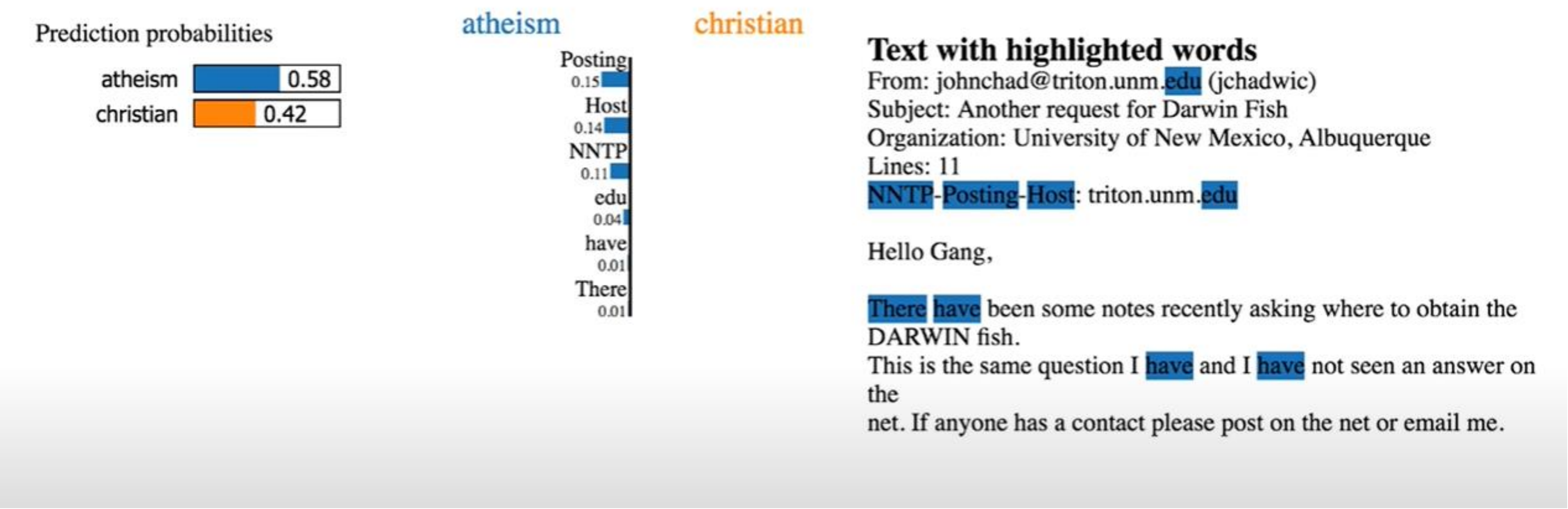
Example

[Code]()



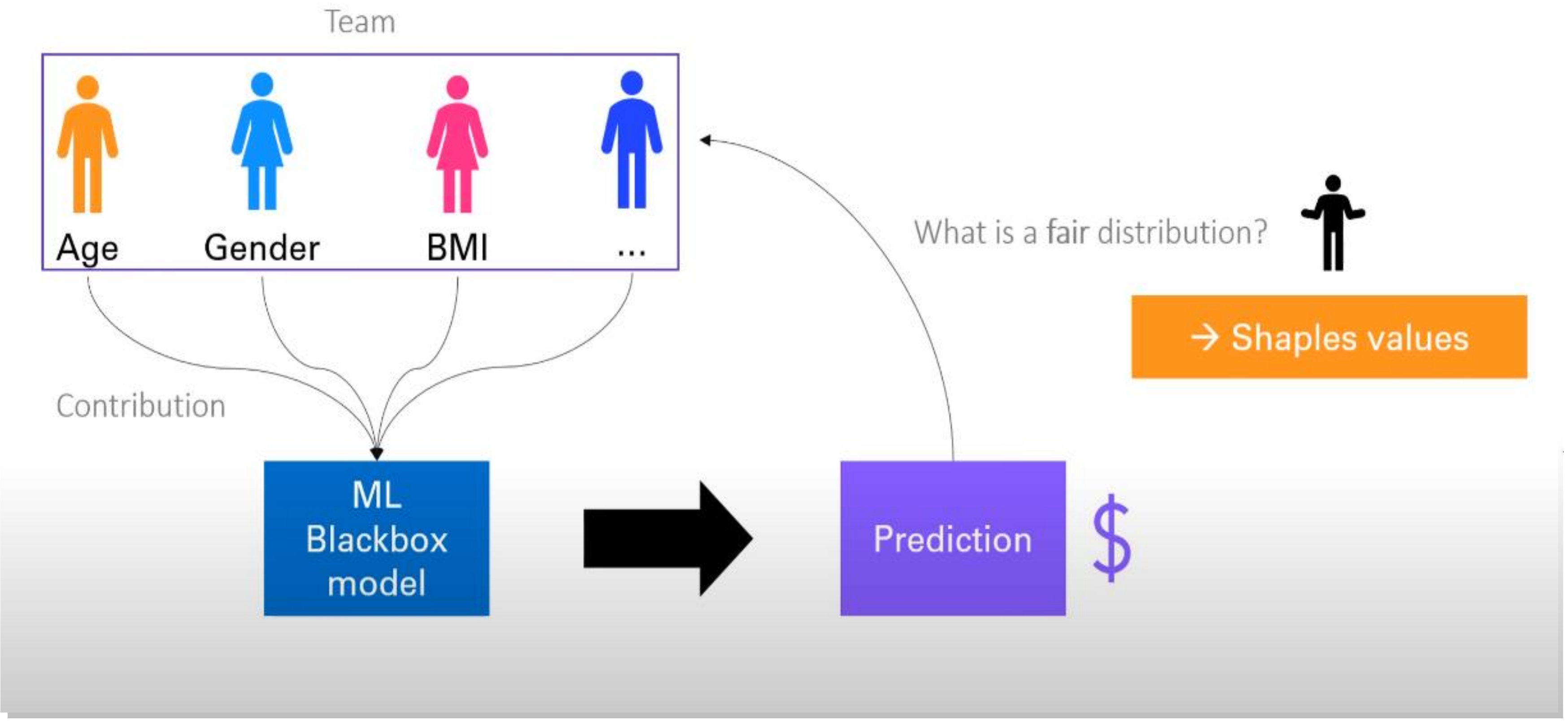
3.2. LIME

Other Examples



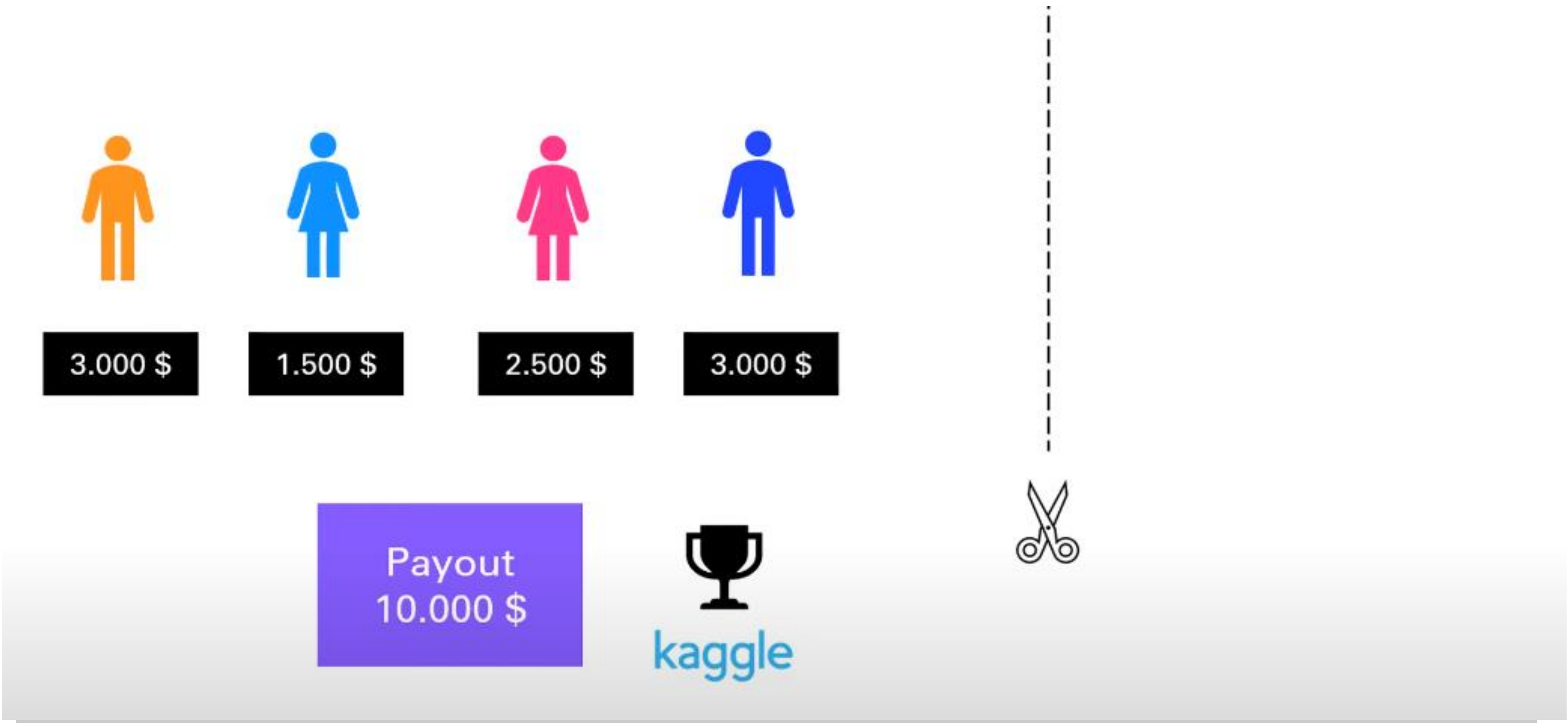
3.3. SHAP

SHapley Additive exPlanations



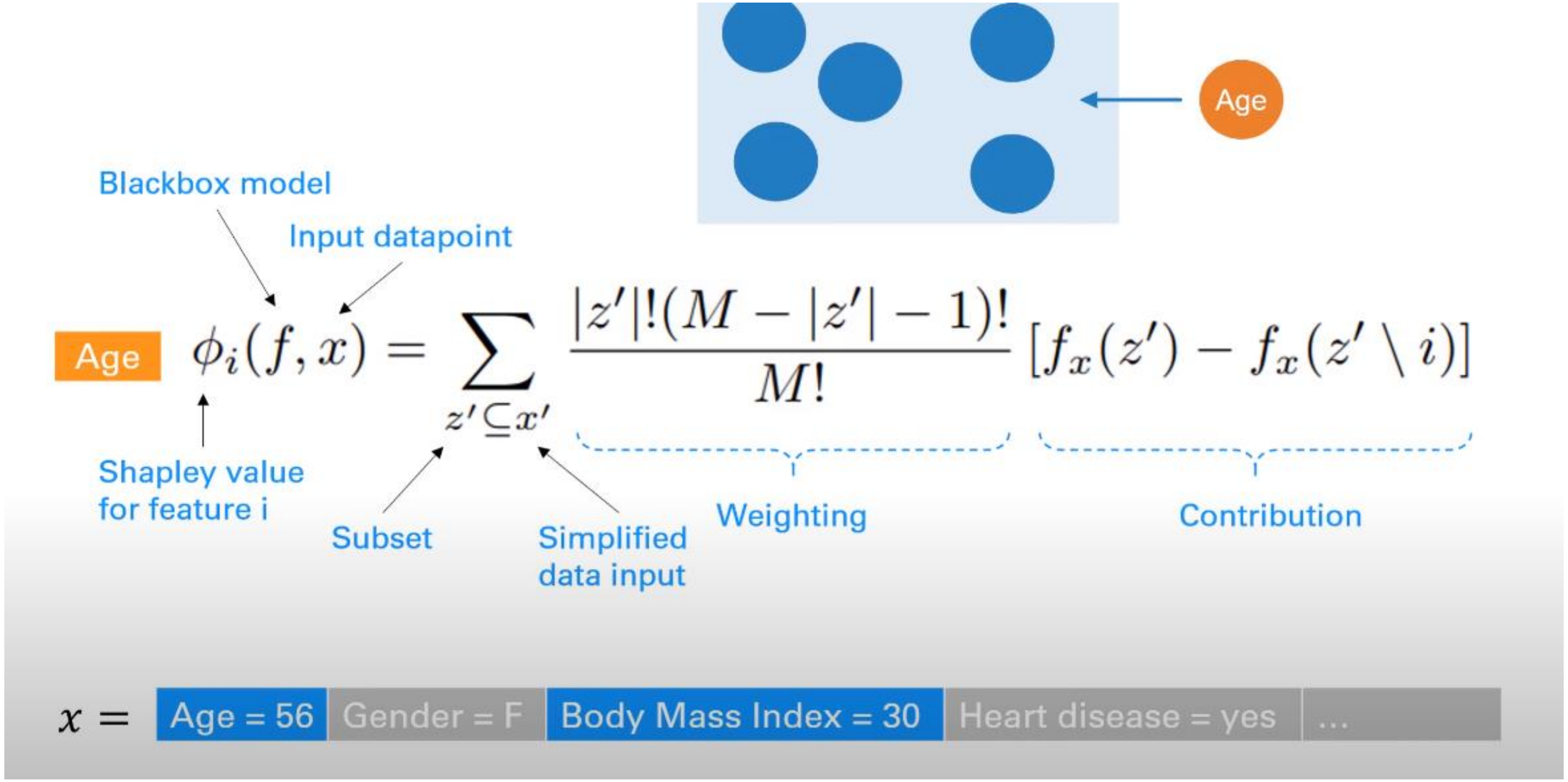
3.3. SHAP

Shaply Values



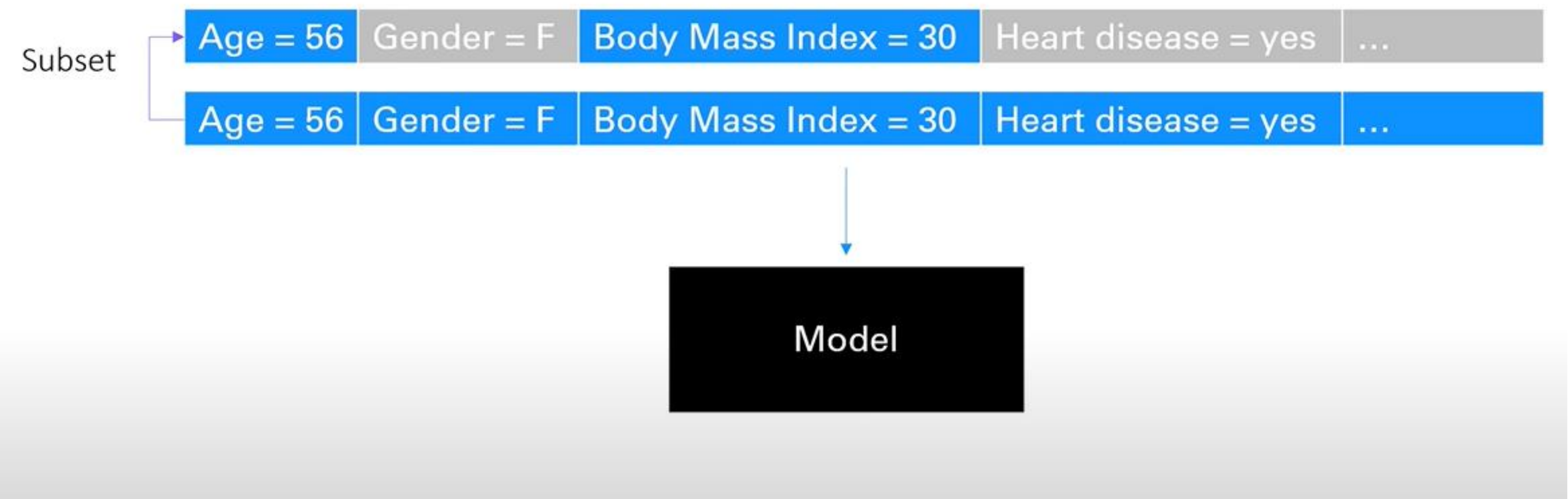
3.3. SHAP

The Math In SHAP



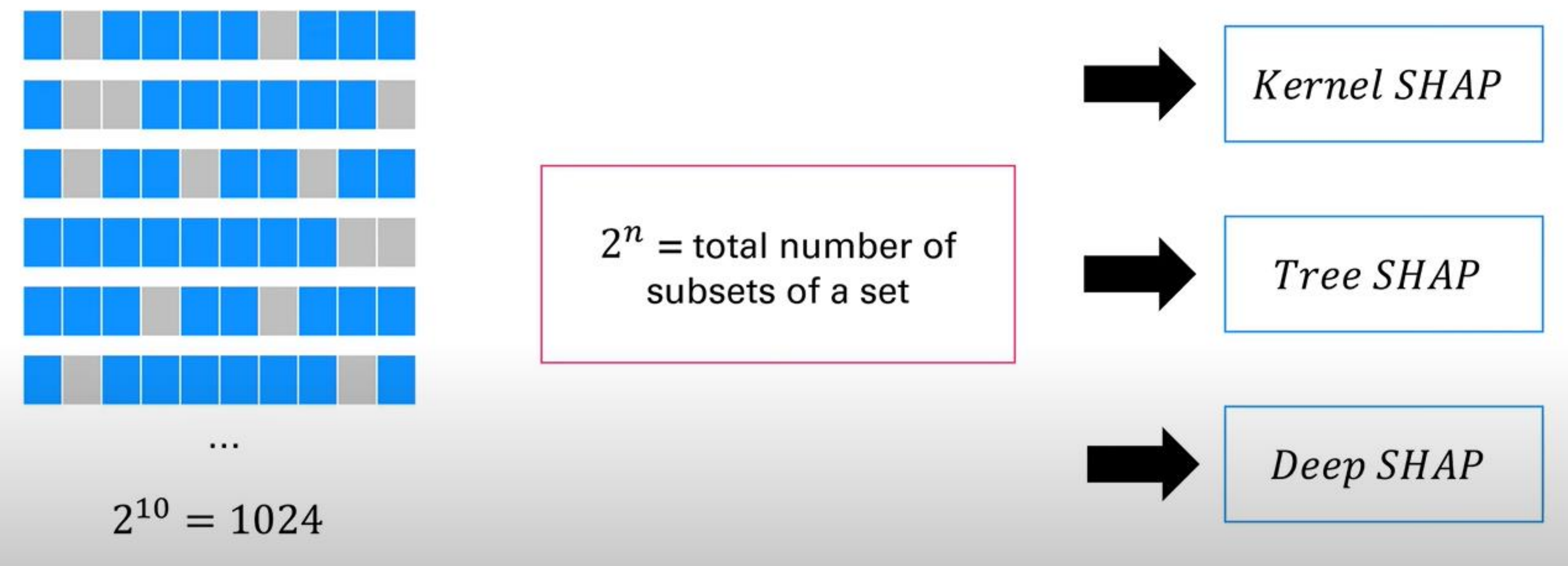
3.3. SHAP

The Math In SHAP



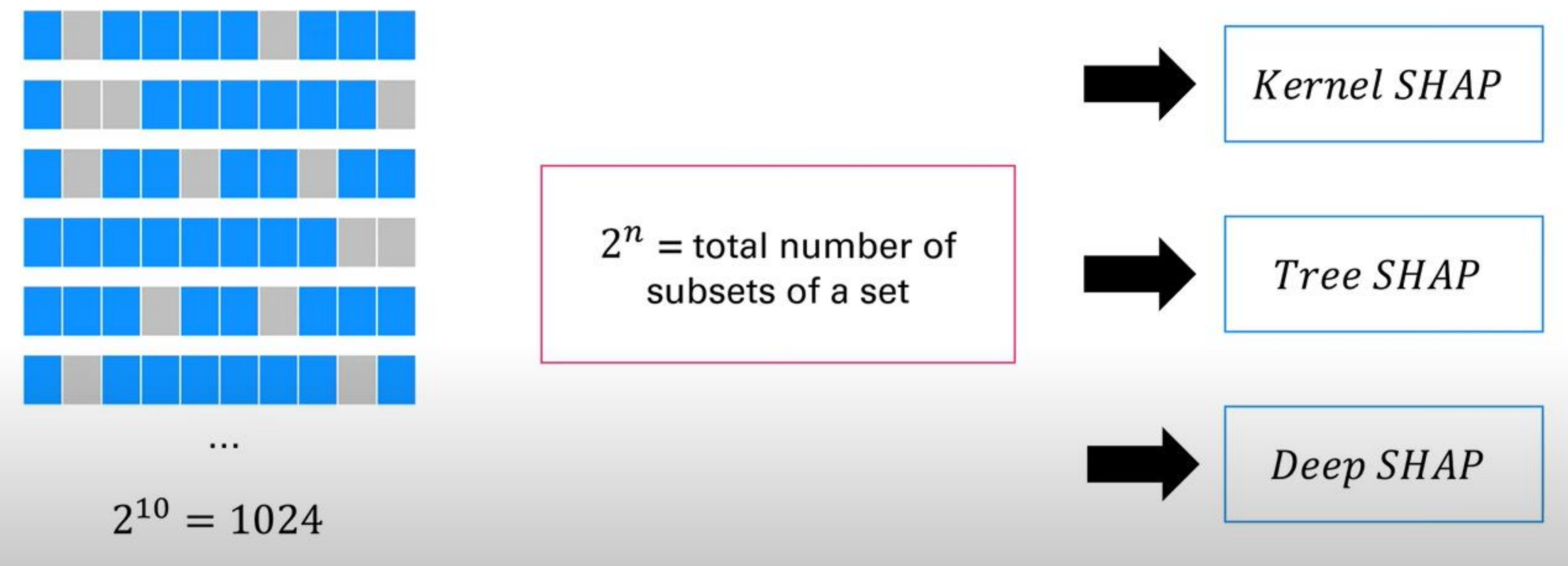
3.3. SHAP

The Math In SHAP



3.3. SHAP

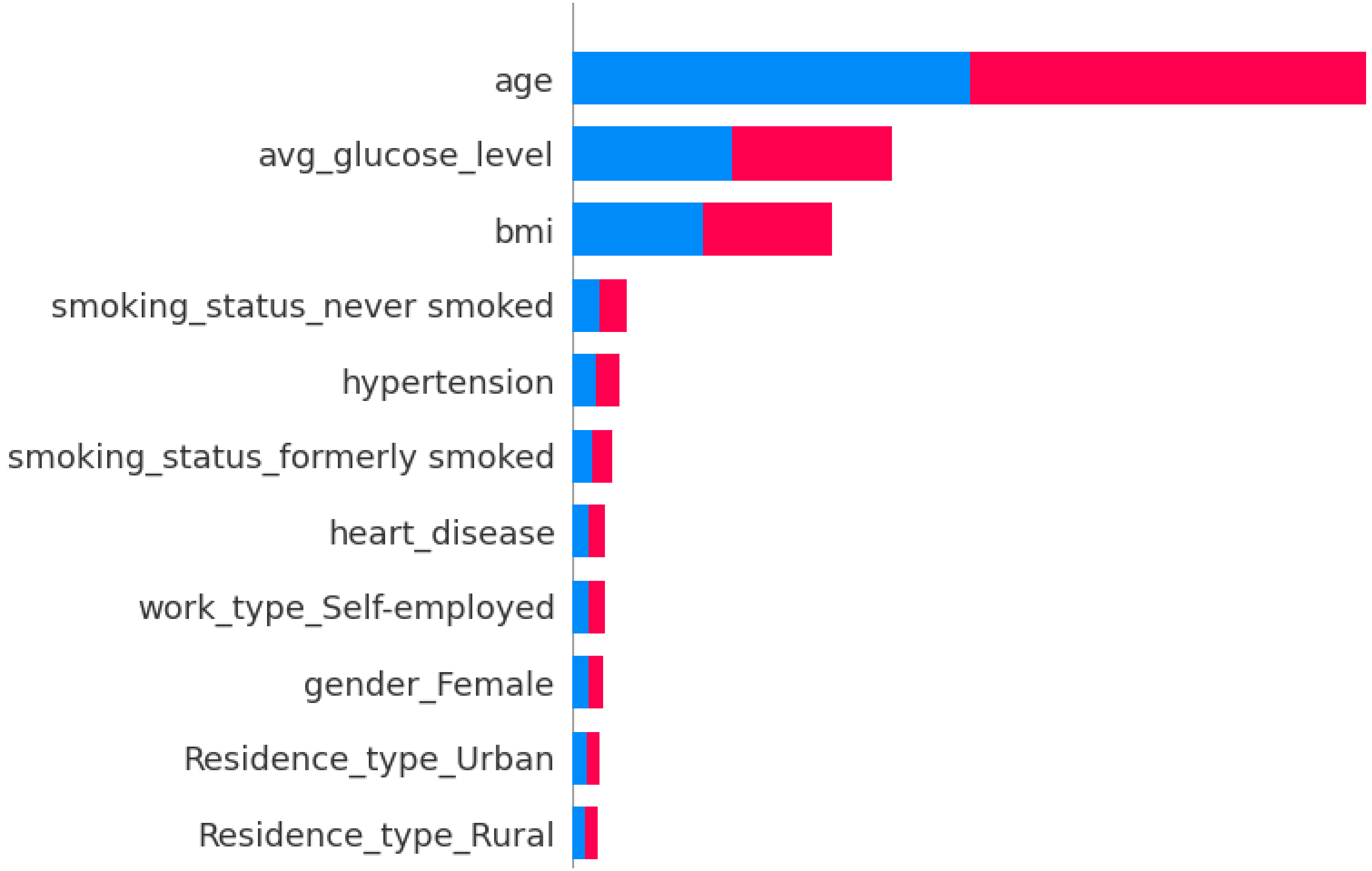
The Math In SHAP



3.3. SHAP

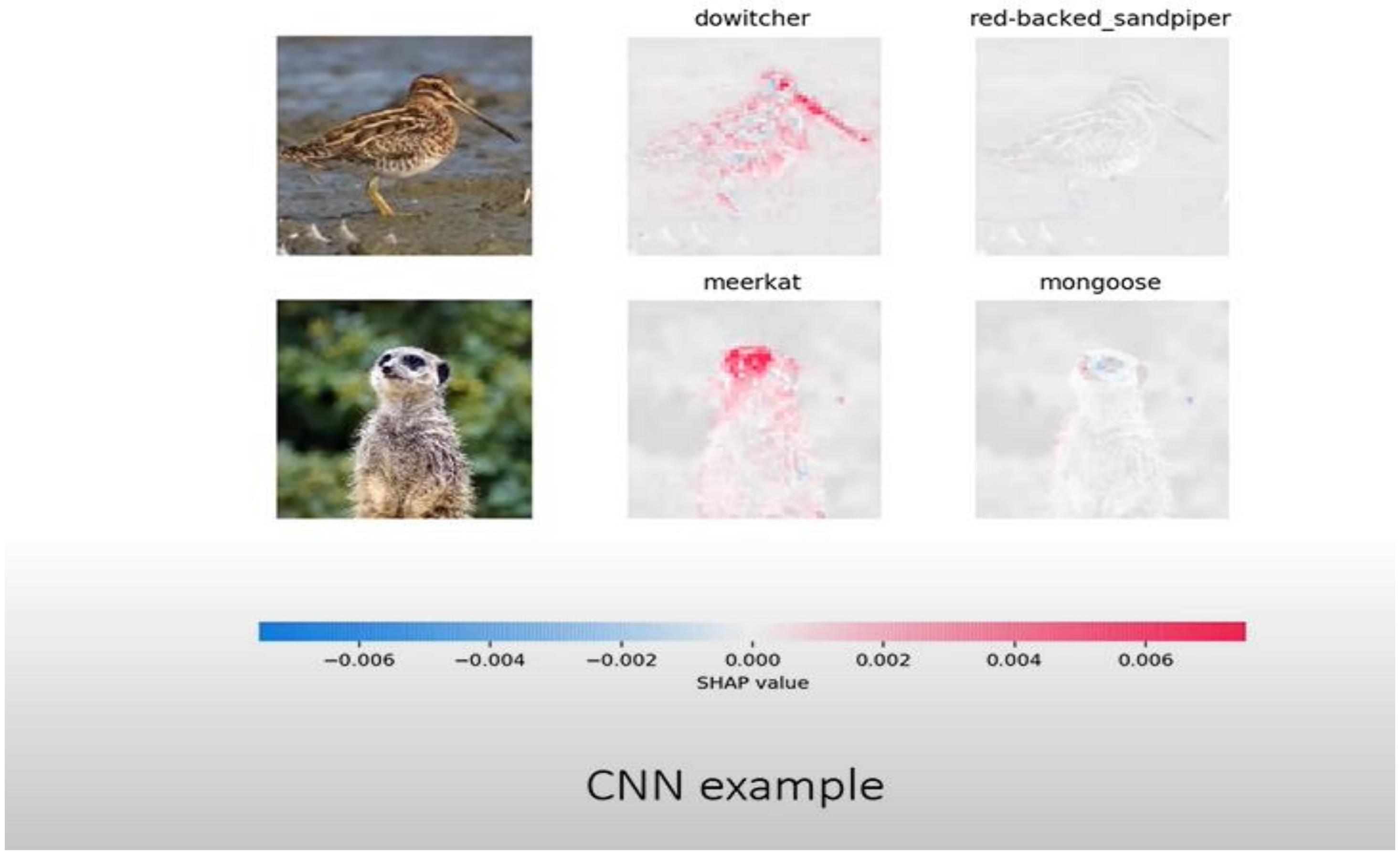
Example

[Code](#)



3.3. SHAP

Other Examples

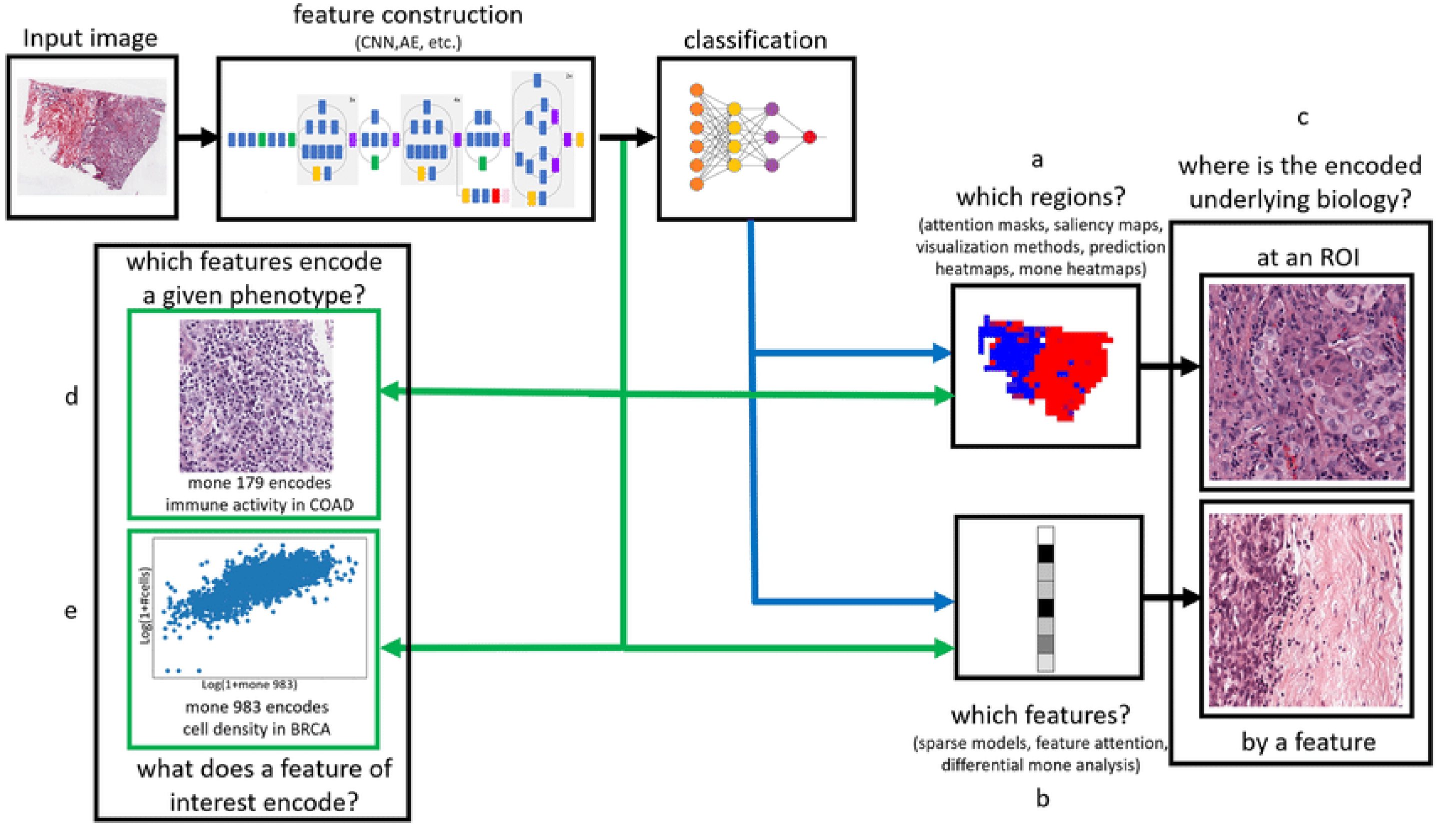


04 Interpretable Deep Learning (Extend)

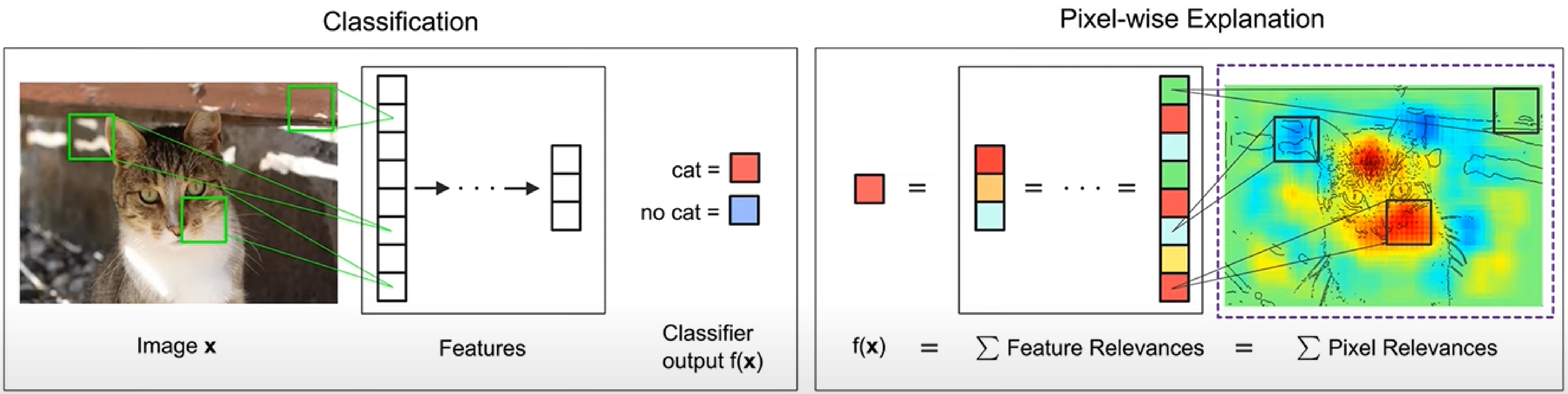
- 4.1 Challenges in interpretability of deep neural networks
- 4.2 Layerwise Relevance Propagation with MRI data
- 4.3 Handle Uncertainty in Deep Learning
- 4.4 Counterfactual explanations and adversarial attacks

4.1. Challenge in interpretability of DNN

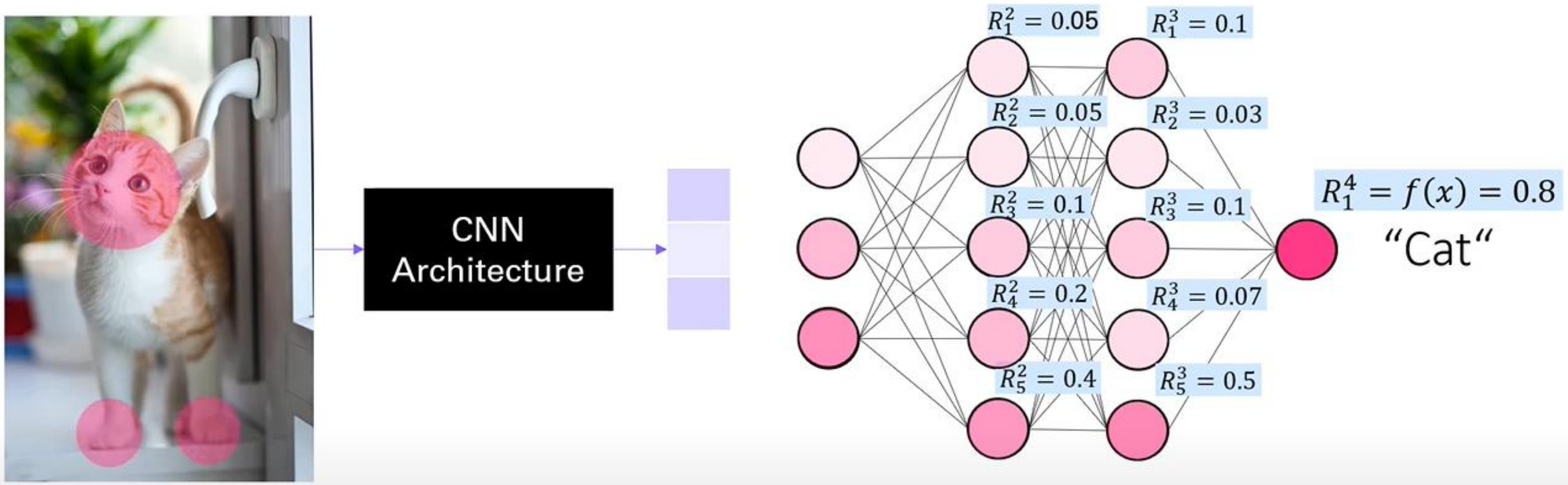
- Black box nature
- High Dimensionality
- Non linear transformation
- Lack of importance feature
- Complexity of Model Architecture



4.2. Layerwise Relevance Propagation with MRI Data

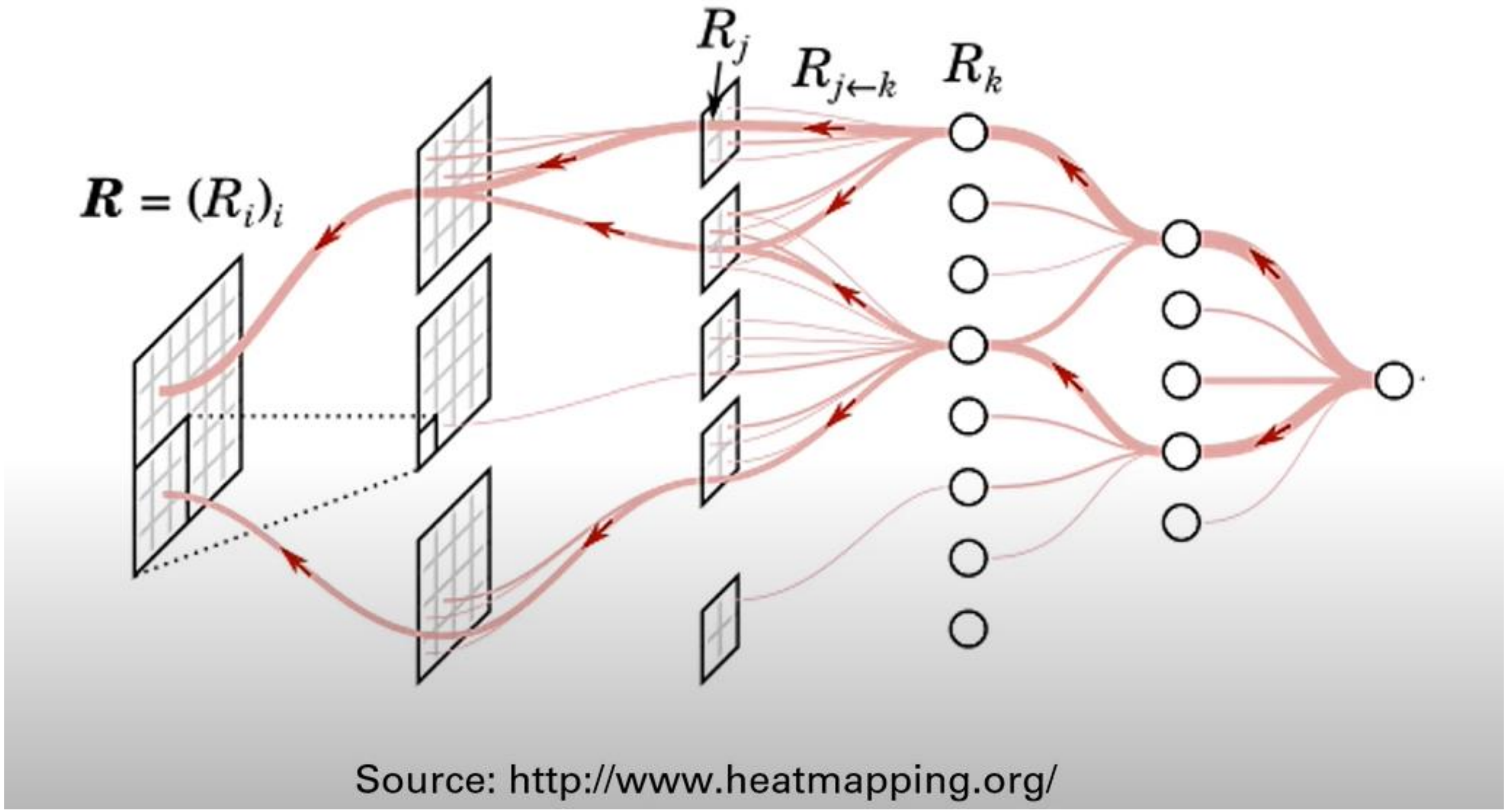


4.2. Layerwise Relevance Propagation with MRI Data

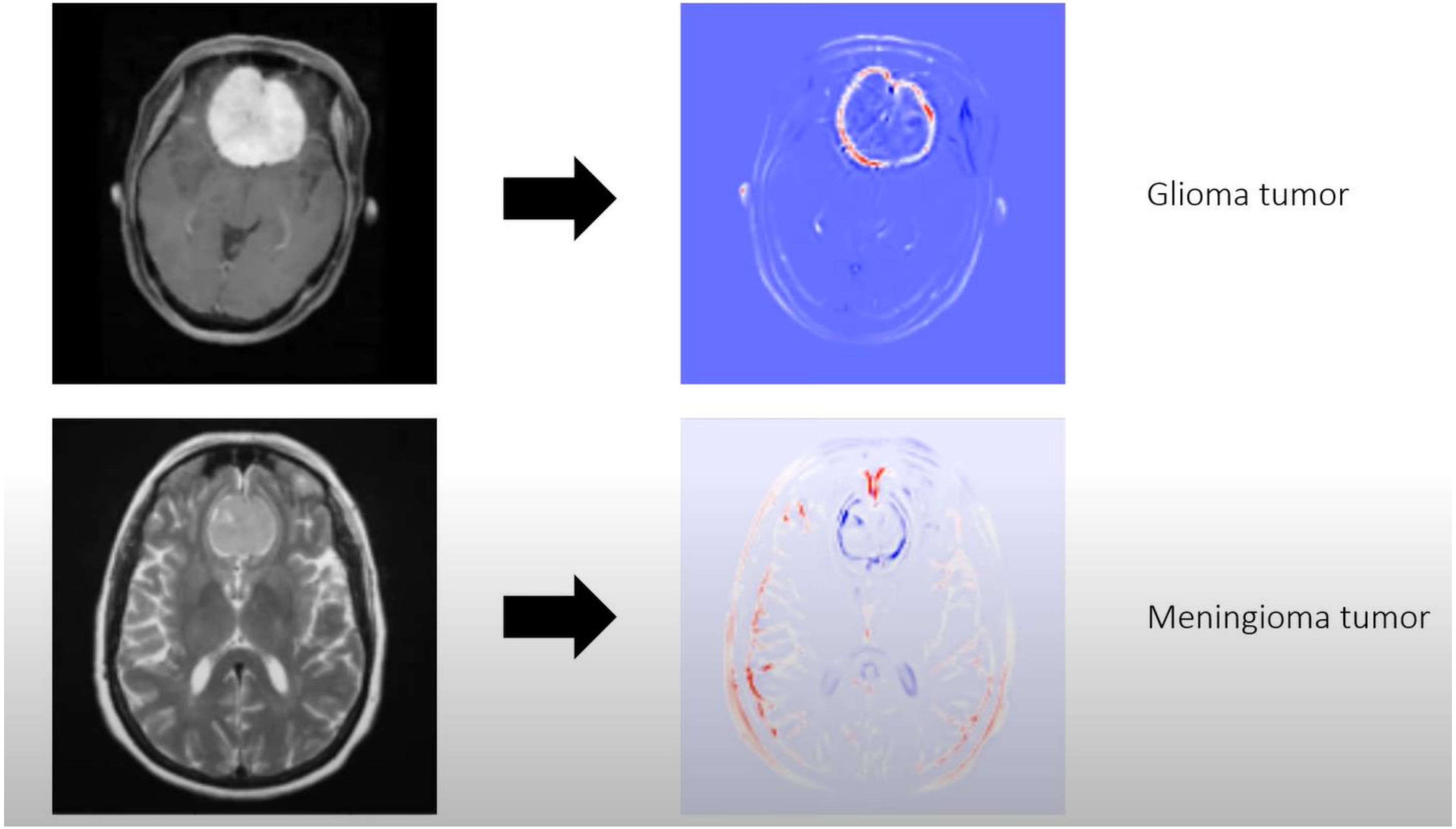


$$R_i^{(l)} = \sum_j \frac{z_{ij}}{\sum_{i'} z_{i'j}} R_j^{(l+1)} \quad \text{with} \quad \mathrm{z_{ij}} = \mathrm{x_i^{(l)} w_{ij}^{(l,l+1)}}$$

4.2. Layerwise Relevance Propagation with MRI Data



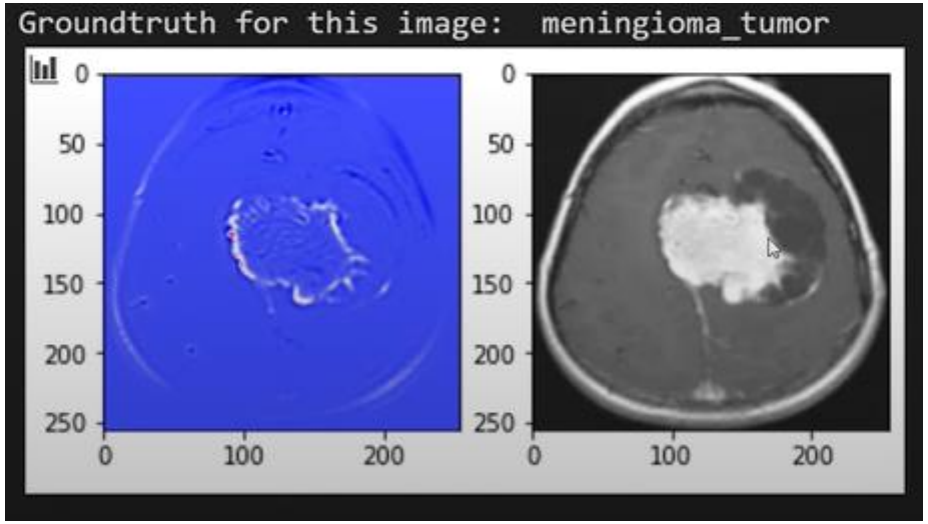
4.2. Layerwise Relevance Propagation with MRI Data

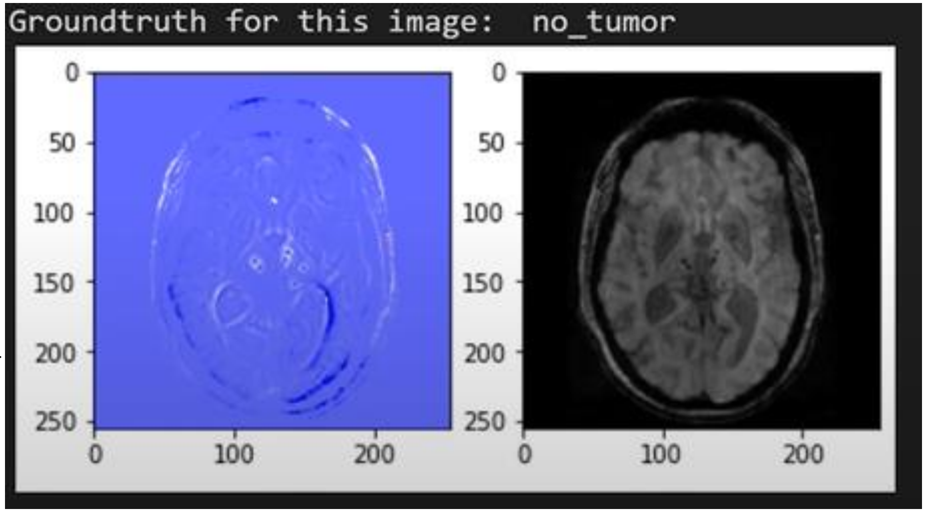


4.2. Layerwise Relevance Propagation with MRI Data

Example

[Code]()

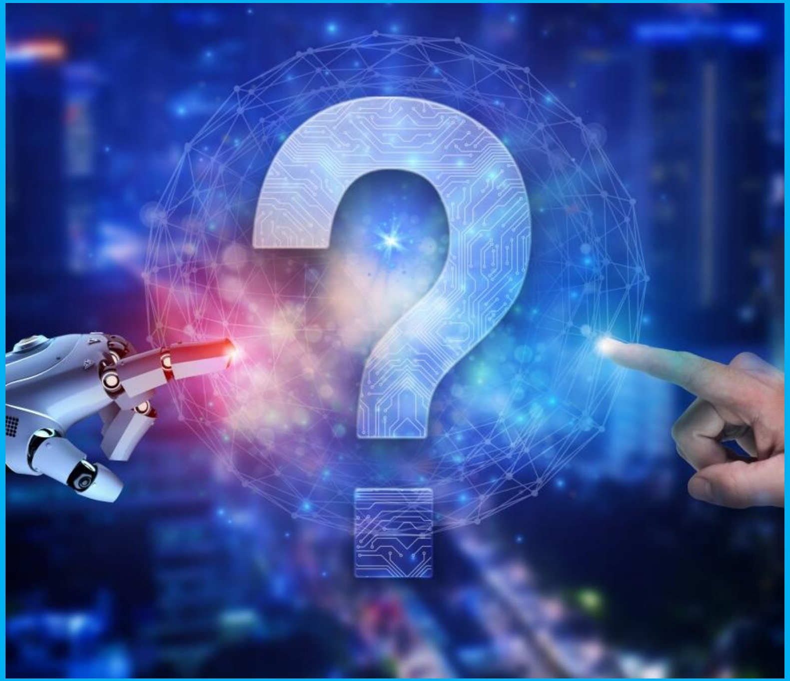




05 Application and Case Studies

- 5.1 Healthcare: Interpretable medical diagnosis systems
- 5.2 Finance: Transparent credit scoring and fraud detection
- 5.3 Law: Explainable legal decision support systems
- 5.4 Social implications and transparency in AI deployment

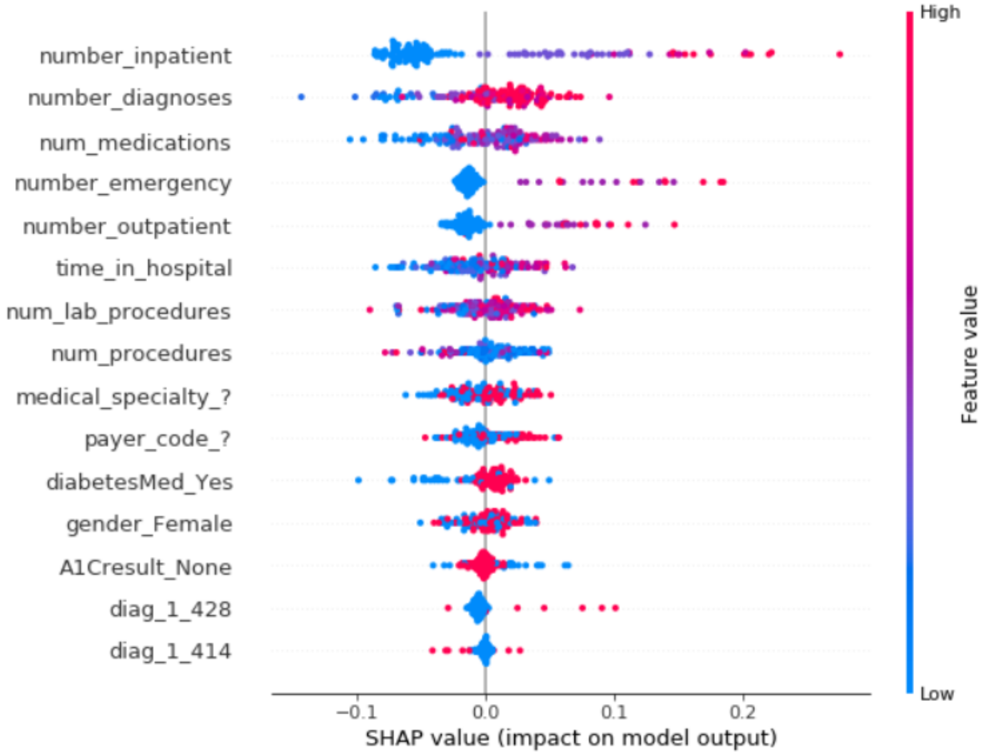
06



Practice

Kaggle Practice

https://www.kaggle.com/code/dansbecker/exercise-advanced-uses-of-shap-values-daily



Appendix – References

1. https://christophm.github.io/interpretable-ml-book/shap.html
2. https://interpret.ml
3. https://arxiv.org/abs/1602.04938
4. https://shap.readthedocs.io/en/latest/

THANK YOU!

Email: contact@tmainnovation.vn

Website: www.tmainnovation.vn